

Laboratory work 3

Task 1 — Generate and insert 200 random rows

QueryQuery History

```
4  INSERT INTO airline
5  (
6      airline_id,
7      airline_code,
8      airline_name,
9      airline_country,
10     created_at,
11     updated_at
12 )
13 SELECT
14     1000 + gs,
15     'AL' || gs,
16     'Random Airline ' || gs,
17     (ARRAY[
18         'Kazakhstan',
19         'Turkey',
20         'France',
21         'Germany',
22         'USA',
23         'Japan',
24         'Brazil',
25         'Poland'
```

Data OutputMessagesNotifications

Showing rows: 1 to 200Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1001	AL1	Random Airline 1	Poland	2024-05-13 18:56:04.400728	2026-09-26 12:17:04.775201
2	1002	AL2	Random Airline 2	Poland	2026-04-21 21:18:04.080482	2026-09-26 12:17:04.775201
3	1003	AL3	Random Airline 3	Brazil	2025-11-17 22:21:34.35826	2026-09-26 12:17:04.775201
4	1004	AL4	Random Airline 4	USA	2024-05-28 10:25:20.09528	2026-09-26 12:17:04.775201

-- TASK 1

-- Generate and insert 200 random rows

```
INSERT INTO airline
(
    airline_id,
    airline_code,
    airline_name,
    airline_country,
    created_at,
    updated_at
)
SELECT
    1000 + gs,
    'AL' || gs,
    'Random Airline ' || gs,
    (ARRAY[
        'Kazakhstan',
        'Turkey',
        'France',
        'Germany',
        'USA',
        'Japan',
        'Brazil',
        'Poland'
    ])[1 + floor(random() * 8)::int],
    NOW() - (random() * INTERVAL '1000 days'),
    NOW()
FROM generate_series(1, 200) AS gs;
```

```
SELECT COUNT(*) AS inserted_random_rows
FROM airline
WHERE airline_id BETWEEN 1001 AND 1200;
```

```
SELECT *
FROM airline
WHERE airline_id BETWEEN 1001 AND 1200
ORDER BY airline_id;
```

Task 2 — Add a new airline named "KazAir" based in "Kazakhstan"

```
-- TASK 2
-- Add KazAir based in Kazakhstan
```

```
INSERT INTO airline
  (airline_id, airline_code, airline_name,
   airline_country, created_at, updated_at)
VALUES
  (1201, 'KZA', 'KazAir',
   'Kazakhstan', NOW(), NOW());
```

```
SELECT *
FROM airline
WHERE airline_name = 'KazAir';
```

QueryQuery History

```
1  -- TASK 2
2  -- Add KazAir based in Kazakhstan
3
4  INSERT INTO airline
5    (airline_id, airline_code, airline_name,
6     airline_country, created_at, updated_at)
7  VALUES
8    (1201, 'KZA', 'KazAir',
9     'Kazakhstan', NOW(), NOW());
10
11 SELECT *
12 FROM airline
13 WHERE airline_name = 'KazAir';
```

Data OutputMessagesNotifications

Showing rows: 1 to 1 of 1

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1201	KZA	KazAir	Kazakhstan	2026-09-26 12:22:01.639434	2026-09-26 12:22:01.639434

Task 3 — Update the airline country "KazAir" to "Turkey"

```
-- TASK 3
-- Update KazAir country to Turkey
```

```
UPDATE airline
SET airline_country = 'Turkey',
    updated_at = NOW()
WHERE airline_name = 'KazAir';
```

```
SELECT *
FROM airline
WHERE airline_name = 'KazAir';
```

Query
Query History

```

1  -- TASK 3
2  -- Update KazAir country to Turkey
3
4  UPDATE airline
5  SET airline_country = 'Turkey',
6     updated_at = NOW()
7  WHERE airline_name = 'KazAir';
8
9  SELECT *
10 FROM airline
11 WHERE airline_name = 'KazAir';

```

Data Output
Messages
Notifications

Showing rows: 1 to 1
Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1201	KZA	KazAir	Turkey	2026-09-26 12:22:01.639434	2026-09-26 12:27:54.489733

Task 4 — Add three airlines at once

-- TASK 4

-- Add three airlines at once

INSERT INTO airline

(airline_id, airline_code, airline_name,
airline_country, created_at, updated_at)

VALUES

(1202, 'AE', 'AirEasy', 'France', NOW(), NOW()),

(1203, 'FH', 'FlyHigh', 'Brazil', NOW(), NOW()),

(1204, 'FF', 'FlyFly', 'Poland', NOW(), NOW());

SELECT *

FROM airline

WHERE airline_name IN ('AirEasy', 'FlyHigh', 'FlyFly')

ORDER BY airline_id;

Query
Query History

```

1  -- TASK 4
2  -- Add three airlines at once
3
4  INSERT INTO airline
5  (airline_id, airline_code, airline_name,
6   airline_country, created_at, updated_at)
7  VALUES
8  (1202, 'AE', 'AirEasy', 'France', NOW(), NOW()),
9   (1203, 'FH', 'FlyHigh', 'Brazil', NOW(), NOW()),
10  (1204, 'FF', 'FlyFly', 'Poland', NOW(), NOW());
11
12  SELECT *
13  FROM airline
14  WHERE airline_name IN ('AirEasy', 'FlyHigh', 'FlyFly')
15  ORDER BY airline_id;

```

Data Output
Messages
Notifications

Showing rows: 1 to 3
Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1202	AE	AirEasy	France	2026-09-26 12:29:22.79583	2026-09-26 12:29:22.79583
2	1203	FH	FlyHigh	Brazil	2026-09-26 12:29:22.79583	2026-09-26 12:29:22.79583
3	1204	FF	FlyFly	Poland	2026-09-26 12:29:22.79583	2026-09-26 12:29:22.79583

Task 5 — Delete all flights whose arrival is in 2024.

```
-- TASK 5
-- Delete all flights whose arrival is in 2024
```

```
DELETE FROM flights
WHERE sch_arrival_time >= '2024-01-01'
AND sch_arrival_time < '2025-01-01'
RETURNING flight_id, sch_arrival_time;
```

Query

Query History

1

-- TASK 5

2

-- Delete all flights whose arrival is in 2024

3

4

DELETE FROM flights

5

WHERE sch_arrival_time >= '2024-01-01'

6

AND sch_arrival_time < '2025-01-01'

7

RETURNING flight_id, sch_arrival_time;

Data Output

Messages

Notifications

≡+

📄

▼

📋

▼

🗑️

🗄️

⬇️

📈

SQL

	flight_id [PK] integer	sch_arrival_time timestamp without time zone
1	1	2024-05-10 12:00:00

Task 6 — Increase the price of all tickets in booking table by 15%

```
-- TASK 6
-- Increase all ticket prices by 15%
```

```
UPDATE booking
SET ticket_price = ticket_price * 1.15,
    updated_at = NOW()
RETURNING booking_id, ticket_price;
```

Query

Query History

1

-- TASK 6

2

-- Increase all ticket prices by 15%

3

4

UPDATE booking

5

SET ticket_price = ticket_price * 1.15,

6

updated_at = NOW()

7

RETURNING booking_id, ticket_price;

Data Output

Messages

Notifications

≡+

📄

▼

📋

▼

🗑️

🗄️

⬇️

📈

SQL

	booking_id [PK] integer	ticket_price numeric (7,2)
1	1	5750.00
2	2	23000.00
3	3	46000.00

Task 7 — Delete all tickets where price is less than 10000

-- TASK 7

-- Delete all tickets where price is less than 10000

```
DELETE FROM booking
WHERE ticket_price < 10000
RETURNING booking_id, ticket_price;
```

Query Query History

```
1  -- TASK 7
2  -- Delete all tickets where price is less than 10000
3
4  DELETE FROM booking
5  WHERE ticket_price < 10000
6  RETURNING booking_id, ticket_price;
```

Data Output Messages Notifications

≡+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL Show

	booking_id [PK] integer	ticket_price numeric (7,2)
1	1	5750.00

Task 8 — Set a default code 'UNK' where airline_code IS NULL.

-- TASK 8

-- Set 'UNK' where airline_code is NULL

```
UPDATE airline
SET airline_code = 'UNK',
    updated_at = NOW()
WHERE airline_code IS NULL
RETURNING airline_id, airline_name, airline_code;
```

Query Query History

```
1  -- TASK 8
2  -- Set 'UNK' where airline_code is NULL
3
4  UPDATE airline
5  SET airline_code = 'UNK',
6      updated_at = NOW()
7  WHERE airline_code IS NULL
8  RETURNING airline_id, airline_name, airline_code;
```

Data Output Messages Notifications

≡+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL

	airline_id [PK] integer	airline_name character varying (50)	airline_code character varying (30)
1	1001	Random Airline 1	UNK

Task 9 — Delete baggage check records created before June 1, 2023, whose result is Not checked.

-- TASK 9

-- Delete unchecked baggage records created before June 1, 2023

```
DELETE FROM baggage_check
WHERE created_at < '2023-06-01'
  AND check_result = 'Not checked'
RETURNING baggage_check_id, check_result, created_at;
```

QueryQuery History

1

-- TASK 9

2

-- Delete unchecked baggage records created before June 1, 2023

3

4

DELETE FROM baggage_check

5

WHERE created_at < '2023-06-01'

6

AND check_result = 'Not checked'

7

RETURNING baggage_check_id, check_result, created_at;

Data OutputMessagesNotifications

≡+

▼

▼

SQL

Showir

	baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone
1	1	Not checked	2023-05-15 10:00:00

Task 10 — Delete airports where state IS NULL and city is Mlawe or Kepuh

-- TASK 10

-- Delete airports where state is NULL

-- and city is Mlawe or Kepuh

```
DELETE FROM airport
WHERE state IS NULL
  AND city IN ('Mlawe', 'Kepuh')
RETURNING airport_id, airport_name, state, city;
```

QueryQuery History

1

-- TASK 10

2

-- Delete airports where state is NULL

3

-- and city is Mlawe or Kepuh

4

5

DELETE FROM airport

6

WHERE state IS NULL

7

AND city IN ('Mlawe', 'Kepuh')

8

RETURNING airport_id, airport_name, state, city;

Data OutputMessagesNotifications

≡+

▼

▼

SQL

Showing ro

	airport_id [PK] integer	airport_name character varying (50)	state character varying (50)	city character varying (50)
1	3	Mlawe Airport	[null]	Mlawe
2	4	Kepuh Airport	[null]	Kepuh

Task 11 — Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at.

-- TASK 11
-- Insert a new unchecked baggage inspection
-- and return generated ID and creation time

```
INSERT INTO baggage_check
(
    check_result,
    created_at,
    updated_at,
    booking_id,
    passenger_id
)
VALUES
(
    'Not checked',
    NOW(),
    NOW(),
    2,
    2
)
RETURNING baggage_check_id, created_at;
```

Query	Query History
1	-- TASK 11
2	-- Insert a new unchecked baggage inspection
3	-- and return generated ID and creation time
4	
5	INSERT INTO baggage_check
6	(
7	check_result,
8	created_at,
9	updated_at,
10	booking_id,
11	passenger_id
12	)
13	VALUES
14	(
15	'Not checked',
16	NOW(),
17	NOW(),
18	2,
19	2
20	)
21	RETURNING baggage_check_id, created_at;

Data Output	Messages	Notifications
	baggage_check_id [PK] integer	created_at timestamp without time zone
1	3	2026-09-26 13:01:15.896293

Task 12 — Convert all country names in the airline table to uppercase

-- TASK 12
-- Convert all airline country names to uppercase

```
UPDATE airline
SET airline_country = UPPER(airline_country),
    updated_at = NOW()
RETURNING airline_id, airline_name, airline_country;
```

Query

Query History

1

-- TASK 12

2

-- Convert all airline country names to uppercase

3

4

UPDATE airline

5

SET airline_country = UPPER(airline_country),

6

updated_at = NOW()

7

RETURNING airline_id, airline_name, airline_country;

Data Output

Messages

Notifications

SQL

	airline_id [PK] integer	airline_name character varying (50)	airline_country character varying (50)
190	1191	Random Airline 191	USA
191	1192	Random Airline 192	BRAZIL
192	1193	Random Airline 193	GERMANY
193	1194	Random Airline 194	USA
194	1195	Random Airline 195	POLAND
195	1196	Random Airline 196	GERMANY
196	1197	Random Airline 197	POLAND
197	1198	Random Airline 198	TURKEY
198	1199	Random Airline 199	BRAZIL
199	1200	Random Airline 200	FRANCE
200	1201	KazAir	TURKEY
201	1202	AirEasy	FRANCE
202	1203	FlyHigh	BRAZIL
203	1204	FlyFly	POLAND
204	1001	Random Airline 1	POLAND

Task 13-Update both the name and country for the airline with airline_id = 5

-- TASK 13

-- Update airline with airline_id = 5

```
UPDATE airline
SET airline_name = 'Global Airways',
    airline_country = 'United Kingdom',
    updated_at = CURRENT_DATE
WHERE airline_id = 5
RETURNING airline_id, airline_name, airline_country, updated_at;
```

Query Query History

```
1  -- TASK 13
2  -- Update airline with airline_id = 5
3
4  UPDATE airline
5  SET airline_name = 'Global Airways',
6     airline_country = 'United Kingdom',
7     updated_at = CURRENT_DATE
8  WHERE airline_id = 5
9  RETURNING airline_id, airline_name, airline_country, updated_at;
```

Data Output Messages Notifications

	airline_id [PK] integer	airline_name character varying (50)	airline_country character varying (50)	updated_at timestamp without time zone
1	5	Global Airways	United Kingdom	2026-09-26 00:00:00

Task 14 — Set state = 'Capital District' for Astana, London and Tokyo

-- TASK 14

-- Set state to Capital District for specific cities

```
UPDATE airport
SET state = 'Capital District',
    updated_at = NOW()
WHERE city IN ('Astana', 'London', 'Tokyo')
RETURNING airport_id, airport_name, city, state;
```

Query Query History

```
1  -- TASK 14
2  -- Set state to Capital District for specific cities
3
4  UPDATE airport
5  SET state = 'Capital District',
6     updated_at = NOW()
7  WHERE city IN ('Astana', 'London', 'Tokyo')
8  RETURNING airport_id, airport_name, city, state;
```

Data Output Messages Notifications

	airport_id [PK] integer	airport_name character varying (50)	city character varying (50)	state character varying (50)
1	2	Astana International Airp...	Astana	Capital District
2	5	London International Airp...	London	Capital District
3	6	Tokyo International Airport	Tokyo	Capital District

Task 15-Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked').

-- TASK 15
-- Mark March 2024 unchecked baggage records as Checked

```
UPDATE baggage_check
SET check_result = 'Checked',
    updated_at = NOW()
WHERE created_at >= '2024-03-01'
    AND created_at < '2024-04-01'
    AND check_result = 'Not checked'
RETURNING baggage_check_id, check_result, created_at;
```

Query

Query History

1

-- TASK 15

2

-- Mark March 2024 unchecked baggage records as Checked

3

4

UPDATE baggage_check

5

SET check_result = 'Checked',

6

updated_at = NOW()

7

WHERE created_at >= '2024-03-01'

8

AND created_at < '2024-04-01'

9

AND check_result = 'Not checked'

10

RETURNING baggage_check_id, check_result, created_at;

Data Output

Messages

Notifications

≡+

▼

▼

SQL

Sh

	baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone
1	10	Checked	2024-03-15 10:00:00